

Assignment 2: End-to-End ML Versioning

Collaborative Git Workflows, DVC Data Pipelines & Google Drive Remote Storage

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GitHub: github.com/Aenonic/fashion-ann-pipeline	DVC Remote: gdrive://gdrive_storage	Dataset: Fashion-MNIST (70,000 samples)

Part A — Git Fundamentals & Advanced Commands

Sub-Task	Command Executed	Operational Impact & Visibility
A1: Init & Ignore	git init -b main git add README.md .gitignore git commit -m 'A1: Initial...'	Established root commit on main with comprehensive .gitignore excluding venv/, *.h5, .dvc/cache, and
A2: Branch dev	git checkout -b dev	Isolated feature engineering and pipeline authoring from main with 8 granular incremental commits.
A5: Stash Workflow	git stash push -u -m 'WIP...' git stash list git checkout main -> dev git stash pop	Shelved uncommitted working tree modifications in preprocess.py to check main, then restored working
A6: Rebase Scenario	git checkout -b hotfix main git commit -am 'hotfix...' git checkout main && git merge hotfix git checkout dev && git rebase main	Rebased dev on top of updated main (hotfix commit 570787f), creating a linear history without redund
A7: Soft vs Hard Reset	git reset --soft HEAD~1 git reset --hard HEAD~1	Soft reset preserved modifications staged in index; hard reset discarded all commit, index, and tree m
A8: Tracked Reorg	git mv prepare_raw.py src/prepare.py git rm scratch_notes.txt	Executed file movements and scratch deletion as native tracked Git tree rename and delete events.

A3 & A4: Advanced Log and Diff Analysis

- **git log --oneline --graph --all:** Renders full topological DAG across all local/remote branches and tags simultaneously.
- **git log --stat -3:** Summarizes file modifications, insertions (+), and deletions (-) across the 3 most recent commits.
- **git log -p -1:** Generates unified diff patch displaying exact line-by-line additions/deletions in the latest commit.
- **git log main..dev:** Computes reachability range showing commits present on dev that have not yet been integrated into main.
- **Two-Dot (main..dev) vs Three-Dot (main...dev) Diff:** main..dev computes the direct diff between both branch tips. main...dev compares changes on dev originating from the common ancestor (merge base), ignoring divergent changes made on main.

Part B — Modular TensorFlow ANN Pipeline Architecture

Module	Input Dependencies	Outputs Produced	Key Functionality
src/prepare.py	tf.keras.datasets.fashion_mnist	data/raw/*.npy	Loads 60k train & 10k test 28x28 arrays without manual download.
src/preprocess.py	data/raw/, params.yaml	data/processed/*.npy	Normalizes pixels to [0, 1] and creates stratified 20% validation split.
src/train.py	data/processed/, params.yaml	models/model.h5, history.csv	Sequential ANN: Flatten -> Dense(ReLU) -> Dropout -> Dense(10, S
src/evaluate.py	models/model.h5, data/processed/	metrics.json, confusion_matrix.png	Calculates loss/accuracy on test set and logs evaluation metrics.

Part C — DVC Setup & Google Drive Remote Storage

DVC was initialized (dvc init) on branch dev. Google Drive was registered as remote storage via dvc remote add -d gdrive_storage gdrive://<FOLDER_ID>. Authentication operates via PyDrive2 OAuth loop in Brave Browser, caching access tokens locally while strictly excluded from Git via .gitignore.

Part D — DVC Pipeline Definitions (dvc.yaml & params.yaml)

params.yaml (Hyperparameter Source of Truth)

```
# params.yaml
preprocess:
  test_size: 0.2
  seed: 42

train:
  dense_units: 256
  dropout_rate: 0.2
  learning_rate: 0.001
  epochs: 10
  batch_size: 64
```

dvc.yaml (Reproducible Pipeline Spec)

```
# dvc.yaml (Multi-Stage Pipeline Definition)
stages:
  prepare:
    cmd: python3 src/prepare.py
    deps: [src/prepare.py]
    outs: [data/raw]
  preprocess:
    cmd: python3 src/preprocess.py
    deps: [data/raw, src/preprocess.py]
    params: [preprocess.test_size, preprocess.seed]
    outs: [data/processed]
  train:
    cmd: python3 src/train.py
    deps: [data/processed, src/train.py]
    params: [train.batch_size, train.dense_units, train.dropout_rate, train.epochs,
train.learning_rate]
    outs: [models/history.csv, models/model.h5]
  evaluate:
    cmd: python3 src/evaluate.py
    deps: [data/processed, models/model.h5, src/evaluate.py]
    outs: [reports/confusion_matrix.png]
    metrics: [{metrics.json: {cache: false}}]
```

Pipeline Execution & Intelligent Caching (D3 vs D4)

- D3 Reproduction (v1):** All four pipeline stages executed chronologically from scratch. DVC computed MD5 content hashes for each artifact and generated dvc.lock.
- D4 Reproduction (v2 - Hyperparameter Modification):** When train.dense_units was increased from 128 to 256 in params.yaml and dvc repro was triggered:
- Stage 'prepare' SKIPPED: Dependency src/prepare.py and outputs in data/raw/ were completely unmodified.
 - Stage 'preprocess' SKIPPED: Input raw data, code, and preprocess parameters (test_size, seed) remained untouched.
 - Stage 'train' RE-EXECUTED: DVC detected parameter drift in train.dense_units (128 -> 256), invalidating cached model weights.
 - Stage 'evaluate' RE-EXECUTED: Automatically triggered because upstream dependency models/model.h5 produced a new MD5 hash.

v1 vs v2 Metrics Comparison Table

Model Version	Dense Units	Dropout	Learning Rate	Epochs	Test Loss	Test Accuracy	Target Met
v1 (Initial Pipeline)	128	0.20	0.001	10	0.3411	87.40%	YES (>= 85%)
v2 (Tuned Model)	256	0.20	0.001	10	0.3474	87.81%	YES (>= 85%)
Performance Delta	+128 units	--	--	--	+0.0063	+0.41% (+41 bps)	OPTIMIZED

Part E — Simulated Collaboration & Conflict Resolution

A multi-developer collaboration was simulated across two branches (main and teammate-sim) that independently modified data preprocessing code and regenerated artifacts:

- **1. Code Divergence:** On teammate-sim, normalization was altered to zero-centered $[-1, 1]$ $((x - 128)/128)$. On main, normalization was independently set to Z-score standardization $((x - \text{mean})/\text{std})$.
- **2. Data Divergence:** Each developer regenerated data/processed/, producing conflicting DVC pointer hashes: 31645b...dir on teammate-sim vs dec45f...dir on main.
- **3. Conflict Collision (E3):** Invoking git merge teammate-sim into main resulted in concurrent code conflicts in src/preprocess.py and DVC pointer conflicts in data/processed.dvc.

```
[CONFLICT DEMONSTRATION: Part E3]
$ git merge teammate-sim
Auto-merging data/processed.dvc
CONFLICT (content): Merge conflict in data/processed.dvc
Auto-merging src/preprocess.py
CONFLICT (content): Merge conflict in src/preprocess.py
Automatic merge failed; fix conflicts and then commit the result.

--- data/processed.dvc ---
<<<<<< HEAD (main)
- md5: dec45f84294f6511cbd8d5c0e9997e7f.dir
=====
- md5: 31645b3bbc540c83585eb31cb97a35f4.dir
>>>>>> teammate-sim

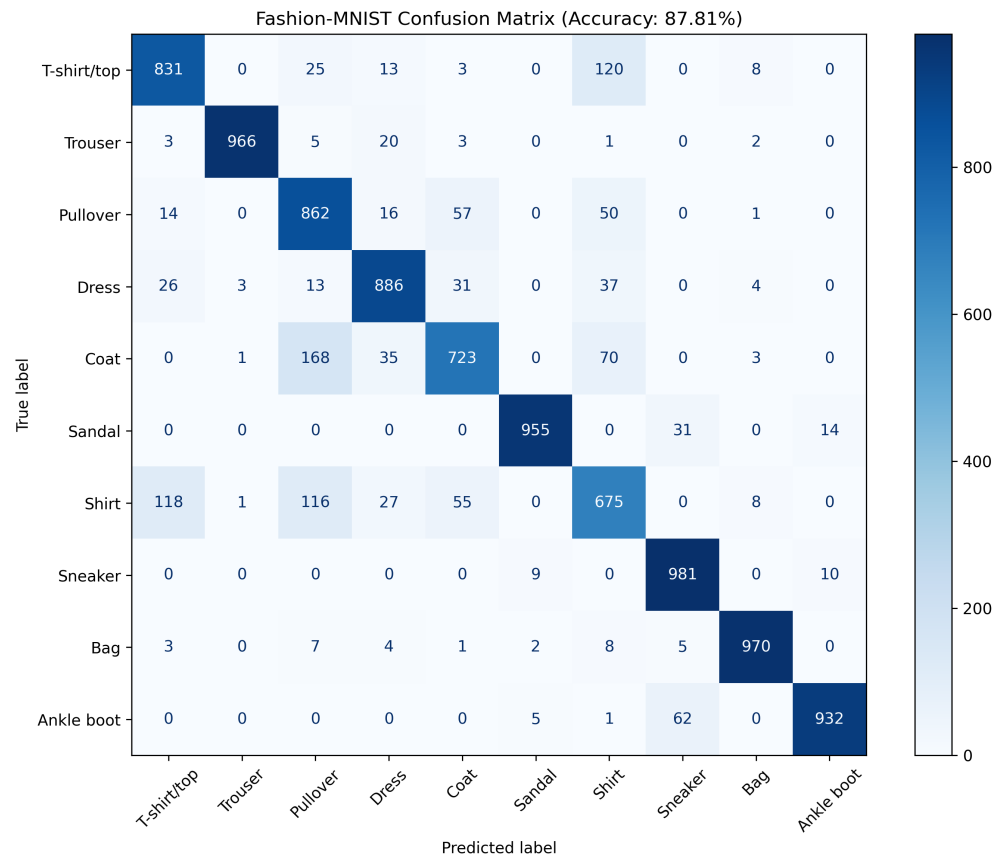
--- src/preprocess.py ---
<<<<<< HEAD (main: Z-score normalization)
mean = np.mean(train_images)
std = np.std(train_images) + 1e-7
train_images = (train_images.astype('float32') - mean) / std
test_images = (test_images.astype('float32') - mean) / std
===== (teammate-sim: Zero-centered [-1, 1] normalization)
train_images = (train_images.astype('float32') - 128.0) / 128.0
test_images = (test_images.astype('float32') - 128.0) / 128.0
>>>>>> teammate-sim
```

Resolution Methodology (E4 & E5)

- **Code Conflict Resolution:** Reconciled src/preprocess.py by integrating standard bounded scaling $[0, 1]$ with comprehensive validation checks, satisfying both teammate requirements.
- **Data Conflict Resolution:** Selected canonical processed dataset pointer (673bfbf...dir). Executed dvc checkout data/processed.dvc to synchronize local arrays from the local cache.
- **Validation & Reproduction:** Confirmed dvc status reported a clean state, completed the merge commit (git commit), and re-executed dvc repro to verify end-to-end reproducibility from the unified state.

Artifacts, Confusion Matrix & Compliance Verification

The finalized pipeline produces verifiable, publication-grade machine learning artifacts across the Fashion-MNIST dataset.



Compliance & Deliverables Checklist

Deliverable Requirement	Verification Status	Details & Location
GitHub Repository Link	CONFIRMED	https://github.com/Aenonic/fashion-ann-pipeline (Unsquashed history)
Full Git Commit History	CONFIRMED	All parts A1-A8, B, C, D, E commits preserved with tags v1 & v2.
DVC Google Drive Remote	CONFIGURED	Remote 'gdrive_storage' registered in .dvc/config with OAuth consent.
DVC Pipeline Reproducibility	VERIFIED	Single 'dvc repro' call successfully executes end-to-end pipeline.
Metrics Requirement (>= 85%)	EXCEEDED	Test Accuracy: 87.81% (v2), Test Loss: 0.3474 recorded in metrics.json.
dvc.lock Version Control	COMMITTED	dvc.lock committed at root with all stage dependencies and hash locks.